

WinAPIReplay

Safety block-list

listed calls are skipped
before the layer decision

Dispatch table

API name $\rightarrow$ executor, $O(1)$

Layer 1 executors

signature-defined;
reconstruct, redirect
12 categories, 237 APIs

Signature
database
per-API

Layer 2 dispatcher

generic; DLL export
lookup, fault-guarded
not counted; not redirected

Argument preparer

type-aware reconstruction
(14 argument types)

Three-tier sandbox

targets rewritten
to disposable places

Run-time state maps

handle · pointer · process-id
updated after every call

Outputs: per-call result record · path transforms · process events

API-call
log
JSON,
ordered
events

emits

emits

reads

updates

uses

uses

reads

uses

not counted; not redirected